# COHEN-MACAULAYNESS WITH LOCALLY NOETHERIAN FIBRES IS FPPF LOCAL ON THE TARGET AND SYNTOMIC LOCAL ON THE SOURCE

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*Editorial note.* Statement span is lines 6101-6107 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6109-6148, the complete author proof (38 body lines).

*Editorial note.* Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Cohen-Macaulay morphisms, fppf descent, and syntomic base change; the reductions are retained.

*Source reference.* Original lemma 036B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6148–6148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02L2: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6148–6148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 036K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6148–6148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 045P: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6148–6148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 038R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6148–6148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *The property $\mathcal{P}(f) =$ "the fibres of $f$ are locally Noetherian and $f$ is Cohen-Macaulay" is local in the fppf topology on the target and local in the syntomic topology on the source.*

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* We have $\mathcal{P}(f) = \mathcal{P}_1(f) \wedge \mathcal{P}_2(f)$ where $\mathcal{P}_1(f) =$"$f$ is flat", and $\mathcal{P}_2(f) =$"the fibres of $f$ are locally Noetherian and Cohen-Macaulay". We know that $\mathcal{P}_1$ is local in the fppf topology on the source and the target, see Descent, Lemmas 02L2 and 036K. Thus we have to deal with $\mathcal{P}_2$.

Let $f : X \to Y$ be a morphism of schemes. Let $\{\varphi_i : Y_i \to Y\}_{i \in I}$ be an fppf covering of $Y$. Denote $f_i : X_i \to Y_i$ the base change of $f$ by $\varphi_i$. Let $i \in I$ and let $y_i \in Y_i$ be a point. Set $y = \varphi_i(y_i)$. Note that

$$X_{i,y_i} = \mathrm{Spec}(\kappa(y_i)) \times_{\mathrm{Spec}(\kappa(y))} X_y.$$

and that $\kappa(y_i)/\kappa(y)$ is a finitely generated field extension. Hence if $X_y$ is locally Noetherian, then $X_{i,y_i}$ is locally Noetherian, see Varieties, Lemma 038R. And if in addition $X_y$ is Cohen-Macaulay, then $X_{i,y_i}$ is Cohen-Macaulay, see Varieties, Lemma 045P. Thus $\mathcal{P}_2$ is fppf local on the target.

Let $\{X_i \to X\}$ be a syntomic covering of $X$. Let $y \in Y$. In this case $\{X_{i,y} \to X_y\}$ is a syntomic covering of the fibre. Hence the locality of $\mathcal{P}_2$ for the syntomic topology on the source follows from Descent, Lemma 036B. Combining the above the lemma follows. □